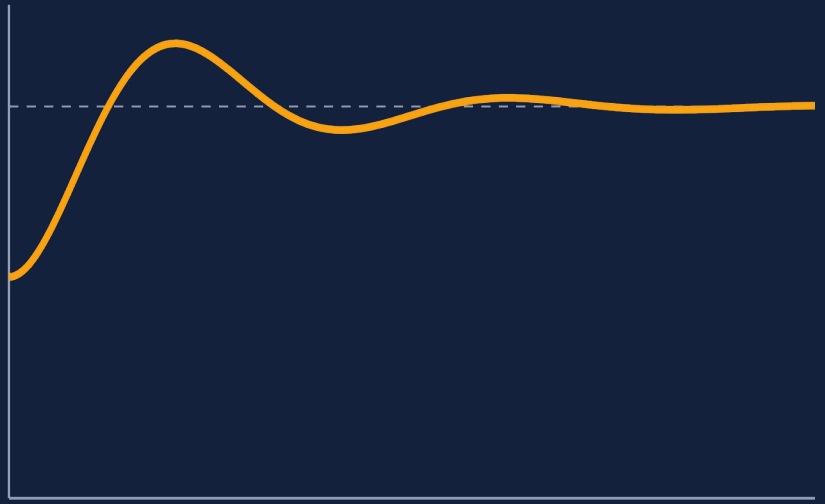
Block Diagrams and Signal Flow Graphs



Control Systems (22EEG53)

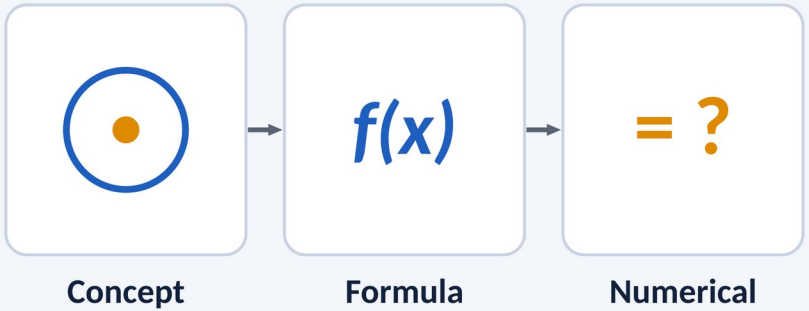
Unit 1: Mathematical Modelling of Physical Systems

Topic: Block diagram reduction, signal flow graphs, Mason's gain formula

Prof. Mitesh Astik

Learning Outcomes

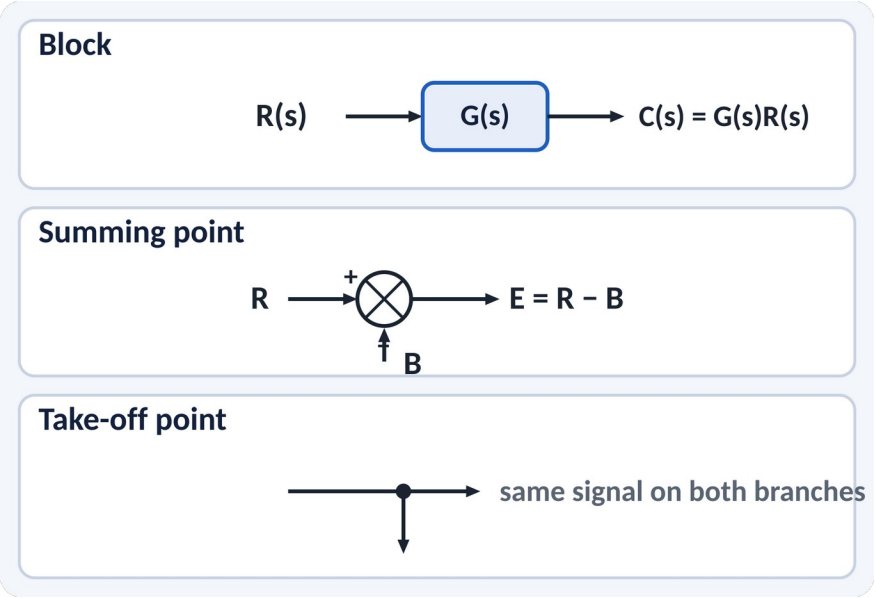
- Reduce block diagrams using standard rules.
- Draw a signal flow graph and identify paths and loops.
- Apply Mason's gain formula to find C/R.
- Compare block reduction with Mason's method.



Linked to CO1
Model dynamical systems by transfer functions.

Recap: What You Already Know

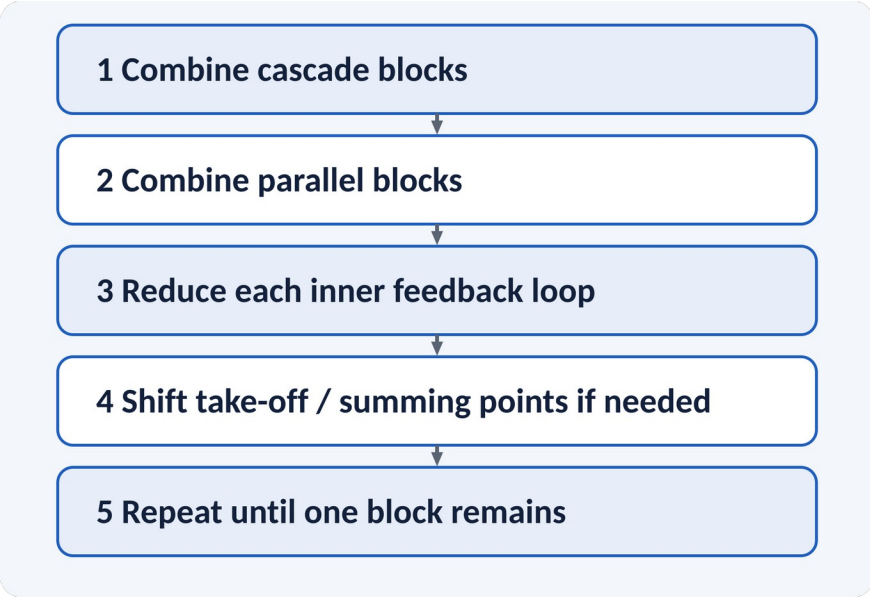
- Transfer function G(s) = C(s)/R(s).
- A block multiplies its input by G(s).
- A summing point adds or subtracts signals.
- A take-off point copies a signal.



C(s) = G(s)·R(s); E(s) = R(s) − B(s)
G(s) transfer function, E error, B feedback signal

Motivation: Many Loops, One Answer

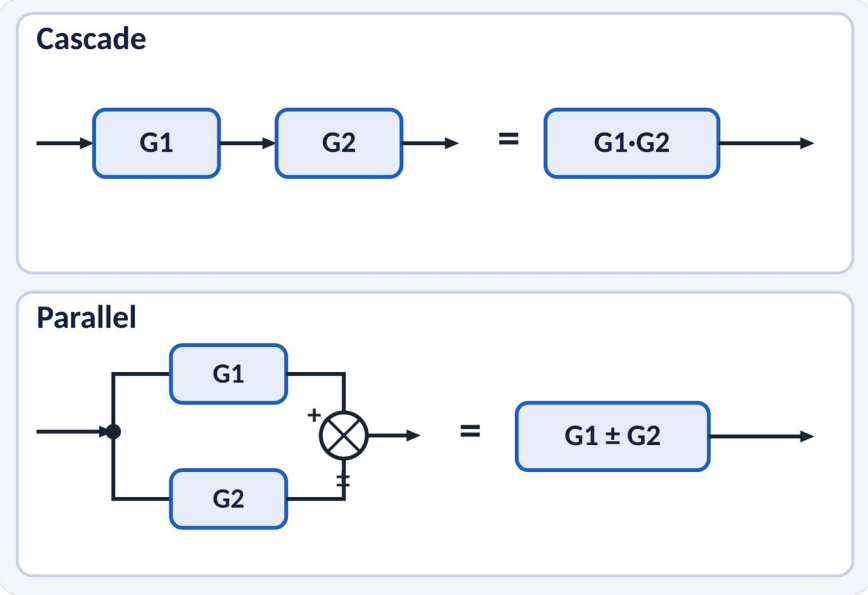
- Real plants have several loops and branches.
- We need one overall transfer function C(s)/R(s).
- Reduction rules simplify step by step.
- Mason's formula gives the answer in one pass.



Goal: replace a complex diagram by a single block T(s) = C(s)/R(s)

Blocks in Cascade and in Parallel

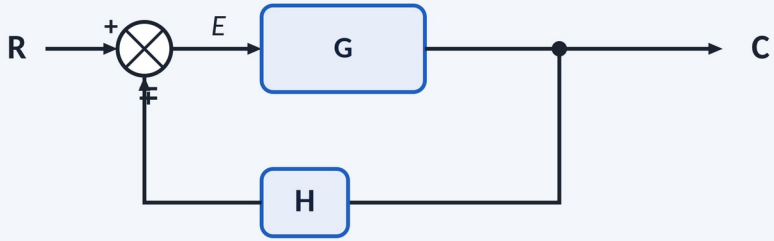
- Cascade blocks multiply: G1·G2.
- Parallel blocks add or subtract: G1 ± G2.
- The sign follows the summing point.
- Order of cascade blocks does not matter (LTI).



Cascade: G = G1 G2; Parallel: G = G1 ± G2

Feedback Loop Reduction

- Forward path G, feedback path H.
- Negative feedback subtracts at the summing point.
- Derive: C = G·E, E = R − H·C.
- Eliminate E to get the closed-loop transfer function.
- Unity feedback means H = 1.



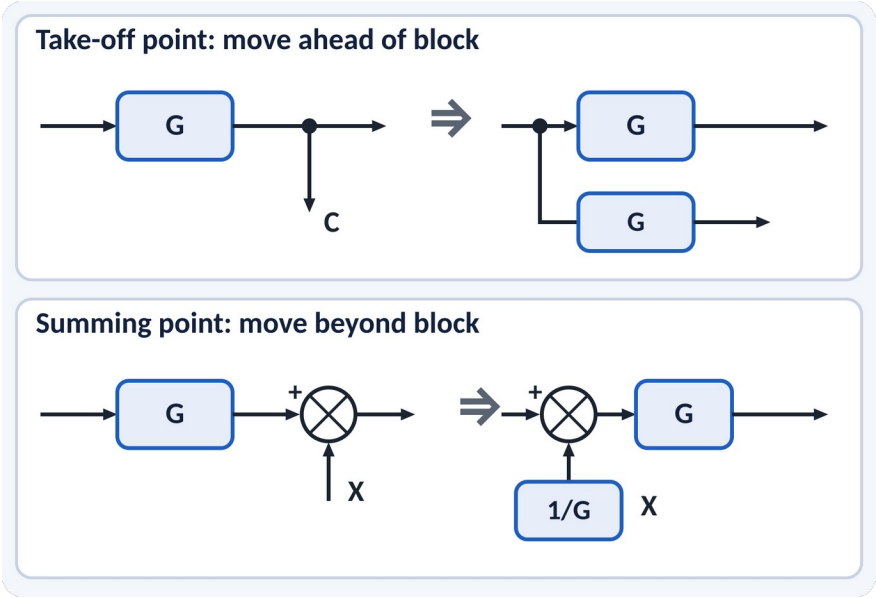
C/R = G / (1 + G·H) (negative feedback)

positive feedback: G / (1 − G·H)

C/R = G/(1 + G·H) (negative); C/R = G/(1 − G·H) (positive)
G forward gain, H feedback gain, GH open-loop gain

Moving Take-off and Summing Points

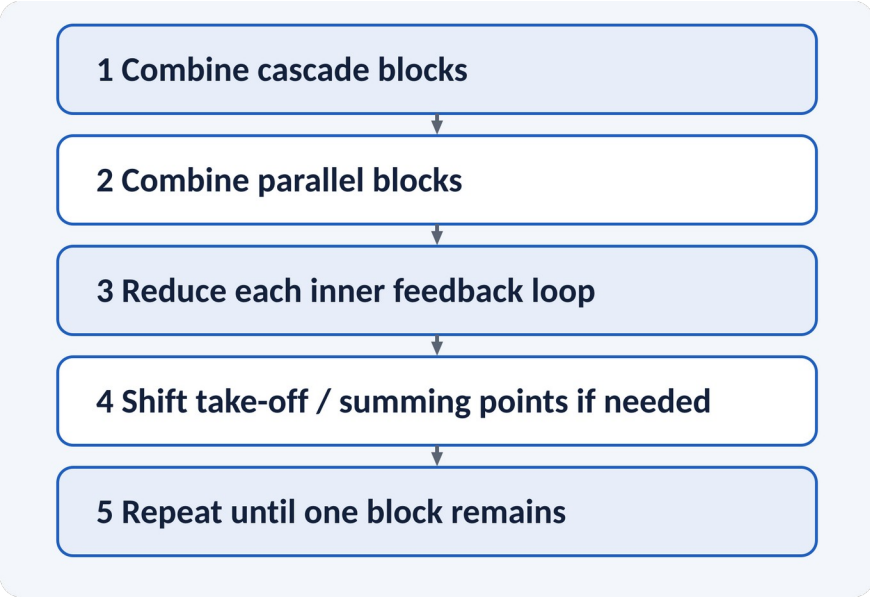
- Shifting is needed when loops overlap.
- Take-off ahead of G: put G in the branch.
- Summing point beyond G: put 1/G in the branch.
- The signals at the outputs must stay unchanged.



Rule: whatever you move across a block G, multiply by G (forward) or 1/G (backward)

Block Diagram Reduction Procedure

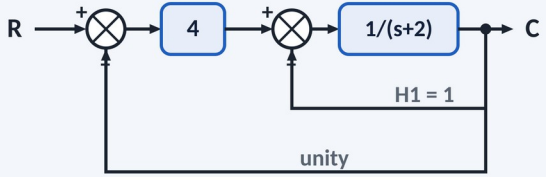
- Combine cascade blocks.
- Combine parallel blocks.
- Reduce inner feedback loops first.
- Shift points to untangle overlapping loops.
- Repeat; finish with the outer loop.



Work from the inside to the outside, one change at a time

Solved Example 1: Block Reduction

- Given the diagram: gain 4, then 1/(s+2) with inner feedback H1 = 1, and outer unity feedback. Find C/R.
- 1. Inner loop: G2/(1 + G2H1) = [1/(s+2)]/[1 + 1/(s+2)] = 1/(s+3).
- 2. Cascade with gain 4: 4/(s+3).
- 3. Outer unity loop: [4/(s+3)]/[1 + 4/(s+3)] = 4/(s+7).



Inner loop: (1/(s+2)) / (1 + 1/(s+2)) = 1/(s+3)

Cascade with 4: 4/(s+3)

Outer loop: C/R = 4/(s+7)

Answer: C(s)/R(s) = 4/(s + 7), a first-order system with pole at s = −7

Signal Flow Graph: Terminology

- Node: a variable; branch: a directed gain.
- Forward path: input to output, no node repeated.
- Loop: closed path, no node repeated.
- Non-touching loops share no node.
- Loop gain: product of branch gains in the loop.



forward paths P1 (blue), P2 (orange); loops L1 (red), L2 (green)

Source node: only outgoing branches; sink node: only incoming branches

Mason's Gain Formula

- One formula gives C/R directly from the graph.
- Δ uses single loops, then pairs of non-touching loops.
- Δk is Δ with loops that touch path k removed.
- Add the contribution of every forward path.

$$T = C/R = \Sigma\, P_k \cdot \Delta_k / \Delta$$

Pk — gain of the k-th forward path

Δ — $1 - \Sigma L + \Sigma L_i L_j - \Sigma L_i L_j L_k \ldots$

Δk — Δ with loops touching Pk removed

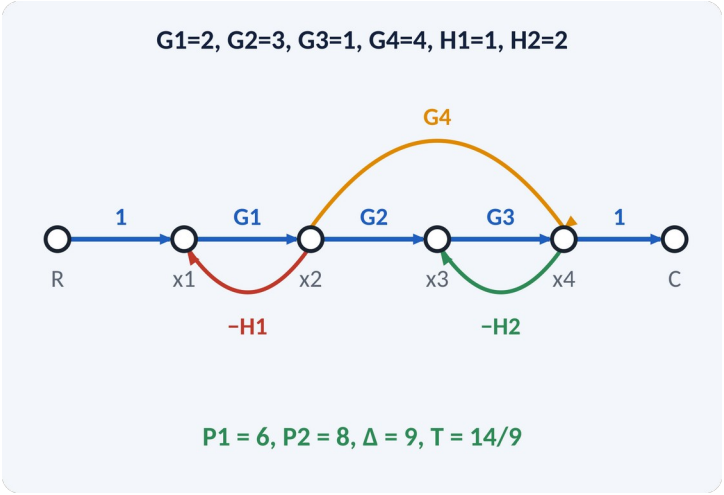
L — loop gain; non-touching = no common node

$T = \Sigma\, P_k \cdot \Delta_k / \Delta; \quad \Delta = 1 - \Sigma L_1 + \Sigma L_2 - \Sigma L_3 + \ldots$

P_k path gain, L_1 single loops, L_2 pairs of non-touching loops

Solved Example 2: Mason's Formula

- Graph with G1 = 2, G2 = 3, G3 = 1, G4 = 4, H1 = 1, H2 = 2 (as drawn). Find C/R.
- 1. Paths: P1 = G1G2G3 = 6; P2 = G1G4 = 8.
- 2. Loops: L1 = −G1H1 = −2; L2 = −G3H2 = −2.
- 3. L1 and L2 are non-touching: L1L2 = 4.
- 4. Δ = 1 − (−2 − 2) + 4 = 9.
- 5. Both paths touch both loops: Δ1 = Δ2 = 1.
- 6. T = (6 + 8)/9 = 14/9.



Answer: C/R = 14/9 ≈ 1.56

Common Mistakes and Exam Tips

Common mistakes

- Missing a forward path or a loop.
- Wrong sign: using plus for negative feedback.
- Calling loops non-touching when they share a node.
- Forgetting to compute Δk for each path.
- Moving a take-off point without compensating.

Exam tips

- Make a table of paths and loops first.
- Write the sign on every loop gain.
- Check DC gain of the answer.
- Redraw after every reduction step.

Summary

- Cascade: G1G2; parallel: G1 ± G2.
- Negative feedback loop: G/(1 + GH).
- Move points across blocks with G or 1/G compensation.
- SFG: nodes, branches, forward paths, loops.
- Mason: T = ΣPkΔk/Δ.

$$T = C/R = \Sigma\, Pk \cdot \Delta k \,/\, \Delta$$

Pk — gain of the k-th forward path

Δ — $1 - \Sigma L + \Sigma L_i L_j - \Sigma L_i L_j L_k \ldots$

Δk — Δ with loops touching Pk removed

L — loop gain; non-touching = no common node

$$T = \Sigma Pk\Delta k/\Delta, \quad \Delta = 1 - \Sigma L_1 + \Sigma L_2 - \ldots$$

Quick Quiz

Q1. Two blocks G1 and G2 in cascade give:
(a) G1+G2 (b) G1G2 (c) G1/G2 (d) G1−G2

Ans: (b)

Q2. Negative feedback loop with forward G, feedback H:
(a) G/(1−GH) (b) G/(1+GH) (c) GH/(1+G) (d) 1/(1+GH)

Ans: (b)

Q3. Moving a summing point beyond a block G needs a block:
(a) G (b) 1/G (c) −G (d) none

Ans: (b)

Q4. Non-touching loops have:
(a) same gain (b) a common node (c) no common node (d) opposite signs

Ans: (c)

Q5. With ΣPkΔk = 14 and Δ = 9, C/R =
(a) 9/14 (b) 1.56 (c) 23 (d) 5

Ans: (b)

References

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   5th Ed., New Age International (P) Ltd., 2008 (Units 1, 2 textbook)

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   5th Ed., Pearson Education Asia / PHI, 2009 (reference)

3. **P. S. Satyanarayana, Concepts of Control Systems**
   2nd Ed., Dynaram Publishers, Bangalore, 2001 (reference)

Next lecture: Unit 2: standard test signals and time response of zero and first-order systems